

JIAXI WANG

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Research CV

Medical Robotics, Embodied AI,
and Electronics Engineering

Last updated July 2026

Research Interests

Autonomous surgical robotics; vision-language-action models; embodied artificial intelligence; medical robotics; computer vision; robot learning; human-computer interaction.

Education

Electronics Engineering	Durham University, University College <i>BEng Electronics Engineering</i>	2023–2027 United Kingdom
	• First Class Honours in Levels 1 and 2; cumulative GPA: 3.72/4.0 . Study in control, signal processing, digital electronics, communications, and electrical engineering.	
Study Abroad	University of Notre Dame <i>Electrical Engineering, International Study Year Abroad</i>	2025–2026 United States
	• Coursework in circuit design, systems theory, semiconductor and optoelectronic devices, power electronics, and computer architecture.	

Research Experience

Summer Undergraduate Research Intern, The Chinese University of Hong Kong Jun–Aug 2026
Medical Robotics Group; Supervisor: Prof. Hongliang Ren

BiliVLA: Scene-Aware VLA for Autonomous ERCP Navigation Supervisor: Jinsong Lin

- **Contributed to a system achieving 96.25% mIoU, 91.96% action precision, and 84.85% navigation success** in ERCP phantom experiments, supporting reliable target understanding and closed-loop control.
- Supported a two-stage pipeline combining grounding-enhanced supervised fine-tuning, GRPO reinforcement optimisation, scene-aware grounding, and safety-aware recovery; integrated the hardware platform, multimodal trajectories, and 3-DoF action definitions.

Teleoperated Flexible Endoscopy Benchmark Supervisor: Jinsong Lin

- **Established a reproducible multimodal benchmark pipeline** and collected an initial **150 synchronised endoscopic video and robot-state segments**, resolving timestamp alignment across video, high-frequency robot states, and PDT hardware parameters.
- Built cleaning, frame-extraction, YOLO ROI-cropping, and 224×224 image-action trajectory workflows; deployed the open-source $\pi_{0.5}$ policy as a baseline for VLA navigation experiments.

Trileg-ESD: Vision-Guided Magnetic Soft Robot Supervisor: Chi Kit Ng

- Advanced a miniature three-legged magnetic soft robot for auxiliary tissue manipulation, structuring controllable motion around squatting, leg lifting, rotation, crawling, and stance-recovery primitives.
- Investigated a vision-language controller mapping current and recent endoscopic frames plus motion instructions to voltage commands for three electromagnetic coils.

Google Summer of Code Mentor 2026, Remote

Open-source neuroinformatics and computational neuroscience

MITACS Globalink Research Intern, McGill University, CIRMMT Jun–Aug 2025

Supervisor: Prof. Suresh Krishna Montreal, Canada

- **Built and evaluated a Python-based, low-latency markerless motion-to-sound workflow** using AI-assisted motion capture and Google MediaPipe for gesture-controlled music generation.
- Contributed to experimental design, coding, analysis, and reporting across engineering, neuroscience, psychology, and human-computer interaction.

HPC Summer Research Student, EPCC, University of Edinburgh Jun–Jul 2024
High-performance computing and scientific workflows United Kingdom

- Applied parallel-computing, large-scale data-processing, and scientific-computing methods to engineering problems after intensive training in HPC systems and architectures.

Publications in submission

EndoWAM: A Grounded World-Action Model for Generalisable Endoscopic Navigation AAAI 2027 conference

Selected Engineering Projects

Automated Material Mover	Durham Engineering 2024–2025
Co-designed and prototyped an autonomous forklift system for industrial material transport under strict hardware constraints, integrating mechanical design, electronics, sensing, and iterative reliability testing. This project got First class honor.	
IMechE Repeatable Vehicle Challenge	Durham University, 2023–2025
Engineered, fabricated, calibrated, and repeatedly tested an autonomous vehicle, improving mechanical repeatability and circuit reliability through iterative debugging.	
Durham Total Artificial Heart	2025
Involved in developing interdisciplinary medical-engineering concepts addressing cardiovascular-technology challenges and translated clinical needs into implementable engineering requirements.	

Technical Skills

Programming	Python; MATLAB; C/C++; Arduino programming; data processing and visualisation.
AI / Robotics	Computer vision; YOLO; vision-language-action models; reinforcement learning; multimodal trajectory datasets; robotic control and sensor integration.
Hardware / Tools	PCB design; analogue/digital electronics; microcontrollers; embedded systems; Multisim; LTspice; Cadence; SolidWorks; Git/GitHub; MediaPipe.
Computing	High-performance computing; parallel architectures; scientific computing; large-scale data processing and technical analysis.

Leadership and Service

Head of Design, DurHack	Sep 2024–2025
Durham University’s annual hackathon	United Kingdom
Led visual identity and user-facing digital communication with software and logistics teams.	
Student Representative and EDI Committee Member	2023–2024
Durham University Department of Engineering	United Kingdom
Represented 150+ engineering students and served as Engineering Society Outreach Executive.	
Co-President, College Ultimate Frisbee	2024–Present
Durham University and University of Notre Dame	UK / US
Managed logistics, recruitment, and multicultural team coordination; competed as a registered USA College player for Notre Dame.	
President, Castle Pride	2023–2024
University College, Durham University	United Kingdom
Led inclusion and community-building programmes and organised student discussions and events.	

Honours and Additional Information

Awards / Service	Two-time finalist, Durham University Annual Art Competition (2023, 2024); Open Day Representative, University College (2023).
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